3.2.4 Free Electrons

The valence electron of different material possesses different energies. The greater the energy of a valence electron, the lesser it is bound to the nucleus. In certain substance, particularly metals, the valence electron possess so much energy that they are very loosely attached to nucleus. These loosely attached valence electron move at random within the material and are called free electrons.

The valence electrons which are very loosely attached to the nucleus are known as free electron.

The free electrons can be easily removed or detached by applying a small amount of external energy. As a matter of fact, these free electrons which determine the electrical conductivity of the material. On the basis of free electron concept, the conductors, insulators and semiconductors can be defined as under:

1. A conductor is a substance which has a large number of free electrons. When potential difference is applied across a conductor, the free electron move towards positive terminal of supply constituting electric current.
2. An insulator is a substance which has practically no free electrons at ordinary temperature. Therefore an insulator does not conduct current under the influence of potential difference.
3. A semiconductor is a substance which has very few free electrons at room temperature. Under the influence of potential difference, a semi-conductor practically conducts no current.

3.3 ELECTRON EMISSION

The liberation of electron from the surface of a substance is known as electron emission.

For electron emission metals are used because they have many free electrons as shown in Figure 3.6. When sufficient energy is given to the free electron, its kinetic energy is increased and thus electron will cross over the surface barrier to leave the metal.

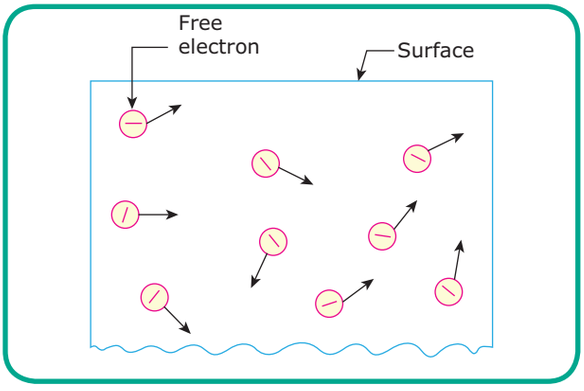


Figure 3.6 Free electron on the surface of the atom

Types of Electron Emission

There are four principle methods of obtaining electron emission from the surface of a metal.

a) Thermal Energy b) Light Energy

c) Secondary Electron d) Electric Field

1. **Thermionic Emission:** In this method, the metal is heated to sufficient temperature (about 2500°C) to enable the electron to leave the metal surface. The number of electrons emitted depends upon the temperature. The higher the temperature, the greater is the emission of electron. This type of the emission is employed in vacuum tubes and CRTS as shown in Figure 3.7.

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